Behavioral Cloning and Reinforcement Learning for Autonomous Driving

Github: adjeielias90

Key:
NN – Neural Network
CNN - Convolutional Neural Network

Abstract

As humans we often learn how to perform tasks via imitation.
By carefully observing others perform a task, then very quickly infer the appropriate actions in the right sequence based on their observations.

In this experiment, we will employ an imitation technique "behavioral cloning from observation".
Our agent will learn and act based on our input, giving us a GIGO(Garbage In, Garbage Out) experience.

Introduction

Behavioral cloning is a method by which human sub-cognitive skills can be learned and reproduced in a program. As the human subject performs the skill, his or her actions are recorded along with the situation that gave rise to the action.

A log of these records is used as input to a neural network. The **NN** outputs a set of rules that reproduce the skilled behavior.
Each time our algorithm makes a mistake, it is slapped with a penalty, and the algorithm performs better after each iteration(epoch). This is known as reinforcement learning.

This method can be used to construct automatic control systems for complex tasks for which classical control theory is inadequate. This is such in our case, which is autonomous driving.
Autonomous driving consists of two main design approaches:

1. end-to-end
2. modular

We will be considering the end-to-end approach for this exercise.

End-To-End Autonomous Driving Design Approach

The end-to-end approach trains a Deep Convolutional Neural Network to generate steering angles based on the input camera observation.
As we know from our introduction, our CNN learns from the behavior of a human expert driver, so this type of learning is called imitation learning.
In the real world, our automaton will navigate by using three cameras instead just one camera. The center camera looks straight, the Left camera looks to the front left and the Right camera looks to the front right.
The task of the autonomous vehicle is to stay in the center of the lane, so by augmenting the Center Camera image data with the Left Camera and Right Camera image data, the vehicle can learn from its own mistakes.

For example:

- If the input image to the CNN is from the Left Camera, then the output of the CNN will instruct the vehicle to steer to the right by a certain amount to stay in the center.

- However if the input image to the CNN is from the Right Camera, then the output of the CNN will instruct the vehicle to steer to the left by a certain amount to stay in the center.

This is known as a stabilizing controller and by doing this we eliminate both the test time trajectory drift and input correlation issues.

An Improvement To Our Approach

To quote Shuri from the famous Black Panther franchise:
“Just because something works doesn't mean that it cannot be improved”.

Our approach, even though works has two major flaws.
1. A human expert driver is required to correctly train the model.
2. Failure cases cannot automatically be dealt with by this approach.

We can mitigate this by further generalizing our learning approach and employing more complex imitation learning techniques such as:

- Apprenticeship Learning by Inverse Reinforcement Learning
- Adversarial Generative Imitation Learning

Recommended Reading/Sources:

- *Behavioral cloning (Michie, Bain, & Hayes-Michie, 1990)*

- *Bagnell, J. A., & Schneider, J. G. (2001). Autonomous helicopter control using reinforcement learning policy search methods. In International conference on robotics and automation, South Korea. IEEE Press, New York.*

- *Bratko, I., & Šuc, D. (2003). Learning qualitative models. AI Magazine, 24(4), 107–119.*

- *Chambers, R. A., & Michie, D. (1969). Man-machine co-operation on a learning task. In R. Parslow, R. Prowse, & R. Elliott-Green (Eds.), Computer graphics: techniques and applications. London: Plenum.*